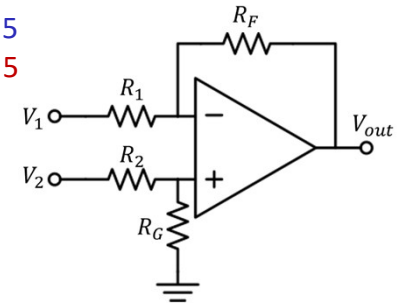
Exercice 5
Aufgabe 5

Solution



$V_+ = V_-$ $\quad R_G = ?$

$$V_{out} = V_2 \frac{R_F}{R_2} - V_1 \frac{R_F}{R_1}$$

$$R_G = R_F \frac{1}{1 + G_1 - G_2}$$

$$V_{out} = V_2 G_2 - V_1 G_1$$

$$V_{out} = \frac{R_1 + R_F}{R_1} \frac{R_G}{R_2 + R_G} V_2 - V_1 \frac{R_F}{R_1}$$

$$G_2 = \frac{R_1 + R_F}{R_1} \frac{R_G}{R_2 + R_G}$$

$$G_2 = (G_1 + 1) \frac{R_G}{R_2 + R_G}$$

$$\frac{R_G}{R_2 + R_G} = \frac{G_2}{G_1 + 1}$$

$$R_G(G_1 + 1) = G_2(R_2 + R_G)$$

$$R_G(G_1 + 1 - G_2) = G_2 R_2$$

$$R_G = R_2 \frac{G_2}{G_1 + 1 - G_2}$$

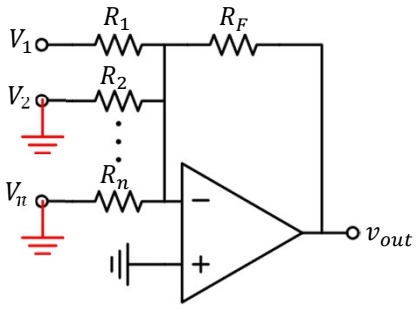
Solution

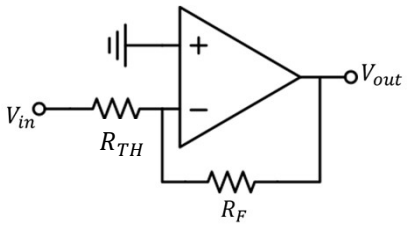
SOL

14

Inverseur sommateur
Invertierender Addierer

Solution





$$v_{out} = -v_{in} \frac{R_F}{R_1}$$

$$R_{TH} = R_1 \parallel \cdots \parallel R_n$$

$$V_{TH_1} = V_1 \frac{R_2 \parallel \cdots \parallel R_n}{R_1 + R_2 \parallel \cdots \parallel R_n}$$

$$V_{TH_1} = V_1 \frac{\frac{1}{\frac{1}{R_2} + \cdots + \frac{1}{R_n}}}{R_1 + \frac{1}{\frac{1}{R_2} + \cdots + \frac{1}{R_n}}}$$

$$V_{TH_1} = V_1 \frac{1}{\frac{R_1}{R_2} + \cdots + \frac{R_1}{R_n} + 1}$$

$$V_{TH_1} = V_1 \frac{1}{R_1 \left(\frac{1}{R_1} + \cdots + \frac{1}{R_n} \right)}$$

$$V_{TH_1} = V_1 \frac{R_1 \parallel \cdots \parallel R_n}{R_1} + \cdots$$

$$V_{out} = -\left(V_1 \frac{R_F}{R_1} + \cdots + V_n \frac{R_F}{R_n} \right)$$

Solution

SOL

16

1